

Week 6

Part B – Track Proposal

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Section 1: Track Chosen

Track D – Neural Operators: DeepONet & Fourier Neural Operator

I selected this track because it merges scientific computing with machine learning and the field of scientific computing is one that interests me the most. The concept of learning an entire solution operator and not just solving one particular PDE instance every single time is something that is both useful and interesting.

Section 2: Project

For this particular project, my objective would be to work on the 1D Burgers' equation with the FNO. The objective is to train an FNO that learns the mapping from an initial condition $u_0(x)$ to the corresponding solution $u(x, T)$ at a fixed time.

I will reproduce the baseline implementation using the `neuraloperator` library and then evaluate its performance on unseen initial conditions. If time permits, I also plan to study how well the trained model generalizes to other viscosities and evaluate how fast it infers compared to the traditional PINN.

The anticipated result would be the FNO model which predicts the solution for new inputs and showcases the main strength of neural operators which is fast inference without re-training for each new equation.

Section 3: Reading Plan

During Week 7, I plan to study the following resources:

- Lu et al., *Learning Nonlinear Operators via DeepONet* (Sections 1–3).
- Li et al., *Fourier Neural Operator for Parametric PDEs*.
- Wang et al., *Learning the Solution Operator of Parametric PDEs with Physics-Informed DeepONets* (Sections 1–3).